

Models Data Extraction & Curation Guidelines

Objective

We are **not building a database of every AI model on the internet**.

For the first version of AIOrbis, our target is:

~2,000 of the best and most relevant AI models globally.

There may be **tens or hundreds of thousands of model variants, fine-tunes, checkpoints and community uploads**. Do not dump them into AIOrbis.

Your job is to **discover broadly, verify carefully, remove duplicates/variants, and select only the strongest models**.

Quality > Quantity.

1. Where to Find Models

Do not randomly search Google or copy models from random model lists.

Priority Source 1 — There's An AI For That

<https://theresanaiforthat.com/models/>

Use this as one of the primary discovery sources for identifying important models across categories.

Priority Source 2 — Models.dev

<https://models.dev/>

Models.dev is particularly useful for structured model metadata such as provider, model specifications, capabilities, pricing, context limits, modalities, release dates and other technical information. Its underlying database is available through an API as well. ([GitHub](#))

Start with these two sources.

Then expand discovery and verification through:

- Official model/provider websites
- Hugging Face
- GitHub
- OpenRouter
- Artificial Analysis
- Official technical documentation
- Major AI research/company announcements
- Credible AI publications

Hugging Face is particularly useful for discovering open-weight models, but **do not treat every Hugging Face model as an AIOrbit candidate**. It contains a huge number of fine-tunes, quantizations, adapters, merges and community uploads. ([Hugging Face](#))

2. What We Actually Want

We want models that are **important, useful, high-quality and currently relevant**.

Prioritize:

- Leading frontier models
- Strong reasoning models
- Strong coding models
- Strong general-purpose LLMs
- High-quality open-weight models
- Image generation models
- Video generation models
- Speech/audio models
- Multimodal models
- Embedding models
- Vision models
- OCR/document models
- Translation models
- Specialized high-performing models
- Important research models
- Models with significant adoption or ecosystem relevance

The model should have a **clear reason to exist in AIOrbit**.

Ask:

"If someone is looking for an AI model, why would they want to discover this model?"

If there is no compelling answer, don't add it.

3. Do NOT Add Every Model Variant

This is one of the most important rules.

A single model family can have:

- Base model
- Instruct model
- Chat version
- Quantized version
- GGUF version
- AWQ version
- GPTQ version
- Fine-tune
- LoRA
- Merge
- Provider-specific version
- Different parameter sizes

Do not treat every variant as a separate important model.

Example

If there is a major model family with:

Model 70B
Model 70B Instruct
Model 70B GGUF
Model 70B GPTQ
Model 70B AWQ

Do not automatically create five AIOrbit models.

Determine whether these are:

different models or simply **different distributions/serving formats of the same underlying model**.

Models.dev itself distinguishes provider-specific serving information from model-level metadata, which is exactly the kind of distinction interns should preserve when deduplicating. ([GitHub](#))

4. Duplicate Rule

Before adding a model, check:

1. Model name
2. Model family
3. Provider/company
4. Official model page
5. Hugging Face repository
6. Model ID
7. Underlying/base model

Same underlying model

→ **One model record**, with relevant providers/versions attached.

Genuinely different model

→ Separate record.

Do not create duplicate records because the same model is available through:

- OpenAI
- OpenRouter
- Together
- Groq
- Fireworks
- Hugging Face
- Other inference providers

Those are **providers**, not necessarily different models.

5. How to Select the Best Models

When multiple models solve the same problem, **compare them rather than adding everything.**

For example, if you discover 50 coding models, ask:

- Which are actually strong at coding?
- Which are actively maintained?
- Which have meaningful adoption?
- Which have credible benchmarks?
- Which are widely used?
- Which have strong real-world performance?
- Which provide something meaningfully different?
- Which are still relevant today?

Then select the strongest ones.

6. Simple Model Quality Score — 100 Points

Use this ranking system:

Factor	Weight
Actual capability / quality	30
Real-world usefulness	20
Adoption / ecosystem traction	15
Benchmark / evaluation performance	10
Activity / current relevance	10
Technical capabilities	5
Documentation / accessibility	5
Differentiation	5
Total	100

Ranking

90–100 — Exceptional

Strongly prioritize.

80–89 — Excellent

Include.

70–79 — Good

Include selectively.

60–69 — Average

Usually skip if stronger alternatives exist.

<60 — Reject.

7. Capability Is the Most Important Factor

Do not rank a model highly simply because it is:

- New
- Popular on social media
- From a famous company
- Open source
- Has many downloads
- Has many GitHub stars

The primary question is:

How good is the model at what it is designed to do?

For example:

Coding model

Evaluate:

- Code generation
- Debugging
- Repository-level tasks
- Tool/function calling
- Reasoning

- Long-context performance

Reasoning model

Evaluate:

- Mathematics
- Logical reasoning
- Complex problem solving
- Planning
- Agentic tasks

Image model

Evaluate:

- Image quality
- Prompt adherence
- Text rendering
- Editing
- Consistency

Speech model

Evaluate:

- Recognition accuracy
- Languages
- Latency
- Voice quality
- Streaming
- Speaker handling

Evaluate models according to their actual purpose.

8. Benchmarks Are Evidence, Not the Final Answer

Use reputable benchmarks when available.

Look at:

- Standard academic benchmarks
- Independent evaluations
- Provider evaluations
- Coding benchmarks
- Reasoning benchmarks
- Multimodal benchmarks
- Human preference evaluations
- Real-world testing

But do **not blindly rank by benchmark leaderboard**.

A model can score extremely well on one benchmark and perform poorly in practical usage.

Therefore:

Benchmark + real-world utility + adoption + technical capability

should determine ranking.

9. Recency and Activity

Models change extremely quickly.

Check:

- Release date
- Latest version
- Last update
- Whether the model is still supported
- Whether the provider recommends it
- Whether newer versions replaced it
- Current API availability
- Current Hugging Face/repository activity

Prefer

New + strong + actively maintained

But don't automatically remove older models.

An older model can remain in AIOrbital if it is:

- Highly capable

- Widely adopted
 - Still supported
 - Important historically/ecosystem-wise
 - Still useful
-

10. Provider Reputation Is a Signal, Not a Guarantee

Models from:

- OpenAI
- Anthropic
- Google
- Meta
- Mistral
- xAI
- DeepSeek
- Qwen
- NVIDIA
- Cohere
- Alibaba
- Microsoft
- etc.

may have strong credibility.

But interns must still evaluate the **individual model**.

Similarly, an independent/open-source model should not be rejected merely because it comes from a smaller organization.

11. Open-Source / Open-Weight Models

Be precise here.

Do not simply write:

"Open Source"

because the model is available on GitHub or Hugging Face.

Check:

- Are weights available?
- What license?
- Are commercial uses permitted?
- Is the code available?
- Are weights available?
- Are there restrictions?
- Is it actually open source or merely open-weight?

Record the exact status where possible.

12. Models.dev Is Excellent for Enrichment

Use Models.dev particularly for structured metadata.

It includes model-level information such as:

- Name
- Family
- Release date
- Last updated
- Reasoning
- Tool calling
- Structured output
- Context limits
- Modalities
- Open weights
- License
- Benchmarks

It also separates provider-specific information such as pricing and provider-specific limits.
([GitHub](#))

Therefore:

Use Models.dev to enrich and cross-check metadata, not as the only source for deciding whether a model is good.

13. What Data to Extract

For each selected model:

Basic

- Model name
- Model family
- Company / creator
- Official website
- Official model page
- Release date
- Last updated
- Model type/category
- Primary task
- Description

Capabilities

- Input modalities
- Output modalities
- Context window
- Maximum output
- Reasoning
- Function/tool calling
- Structured output
- Vision
- Audio
- Multimodal capability
- Embeddings, where applicable

Deployment

- API available
- Providers
- Platforms
- Open-weight/open-source status
- License
- Hugging Face URL
- GitHub URL
- Official documentation

Commercial

- Pricing
- Free/freemium/paid
- Input cost
- Output cost
- Other relevant pricing

Quality / ranking

- Relevant benchmark results
 - Adoption signals
 - Downloads/usage where available
 - AIOrbit quality score
 - Last verified date
 - Source(s)
-

14. Do Not Collect Random Data

Only collect information that helps users:

discover → understand → compare → evaluate → use the model.

Do not scrape irrelevant information simply because it exists on the model's website.

For individual creators, collect only relevant **public professional attribution** when needed.

Do not collect:

- Personal phone numbers
 - Home addresses
 - Private emails
 - Personal IDs
 - Family information
 - Unrelated personal social accounts
 - Other unnecessary personal information
-

15. Final Selection Process

The practical workflow for interns is:

Discover

There's An AI For That + Models.dev

↓

Expand

Official sources + Hugging Face + GitHub + other credible model sources

↓

Remove

Duplicates + fine-tune spam + quantization duplicates + abandoned models + irrelevant models

↓

Verify

Official model page / provider documentation

↓

Evaluate

Capability + usefulness + benchmarks + adoption + activity

↓

Score

0–100

↓

Compare

When several models perform the same task, keep the strongest ones.

↓

Final database

Maximum ~2,000 high-quality models

Golden Rule

We are not building the biggest model database. We are building the model database people can trust.

If you discover 20,000 models and only 2,000 deserve to be there, keep the 2,000.

If only 1,500 meet the quality bar, keep 1,500.

Never lower the quality bar just to reach 2,000.

Discover broadly → verify from authoritative sources → eliminate variants/duplicates → rank by actual capability → curate the best models.